

DSA Exam Practice Set

Practice set based on the DSA topics and problem patterns covered in class.

Instructions: Attempt each question independently first. Focus on the approach, edge cases, and time/space complexity.

Arrays

Q1. Find Largest and Smallest

Given an integer array, find the smallest and largest element.

Example: [5,2,9,1,7]

Expected Output: Smallest = 1, Largest = 9

Constraints: $1 \leq n \leq 10^5$; $-10^9 \leq \text{arr}[i] \leq 10^9$

Pattern/Topic: Array Traversal

Q2. Second Largest Element

Find the second largest distinct element without sorting the array. Assume at least two distinct values exist.

Example: [10,5,8,10,3]

Expected Output: 8

Constraints: $2 \leq n \leq 10^5$

Pattern/Topic: Array Traversal

Q3. Move Zeroes

Move all zeroes to the end while keeping the relative order of non-zero elements. Modify the array in-place.

Example: [0,1,0,3,12]

Expected Output: [1,3,12,0,0]

Constraints: $1 \leq n \leq 10^5$

Pattern/Topic: Two Pointer

Q4. Remove Duplicates from Sorted Array

Remove duplicates from a sorted array in-place so each distinct value appears once. Return the number of unique values.

Example: [1,1,2,2,3]

Expected Output: 3

Constraints: $1 \leq n \leq 10^5$

Pattern/Topic: Two Pointer

Q5. Missing Number

An array contains n distinct numbers from 0 to n . Find the missing number.

Example: [3,0,1]

Expected Output: 2

Constraints: $1 \leq n \leq 10^5$

Pattern/Topic: XOR / Array

Q6. Single Number

Every element appears twice except one. Find the element that appears once.

Example: [4,1,2,1,2]

Expected Output: 4

Constraints: $1 \leq n \leq 10^5$

Pattern/Topic: XOR

Q7. Maximum Subarray Sum

Find the maximum sum of any contiguous subarray.

Example: [-2,1,-3,4,-1,2,1,-5,4]

Expected Output: 6

Constraints: $1 \leq n \leq 10^5$

Pattern/Topic: Kadane's Algorithm

Q8. Longest Consecutive Sequence

You are given an **unsorted integer array**.

Your task is to find the **length of the longest sequence of consecutive integers** that can be formed using the elements of the array.

A sequence is called **consecutive** if every next number is exactly **1 greater** than the previous number.

Aim for $O(n)$ average time.

Example: [100,4,200,1,3,2]

Expected Output: 4

Constraints: $0 \leq n \leq 10^5$

Pattern/Topic: HashSet

Q9. Two Sum

Return indices of two different elements whose sum equals target. Assume exactly one solution.

Example: nums=[2,7,11,15], target=9

Expected Output: [0,1]

Constraints: $2 \leq n \leq 10^5$

Pattern/Topic: HashMap

Q10. Majority Element

Find the element appearing more than $n/2$ times. A majority element is guaranteed.

Example: [2,2,1,1,1,2,2]

Expected Output: 2

Constraints: $1 \leq n \leq 10^5$

Q11. Maximum Product Subarray

Find the contiguous subarray having the largest product.

Example: [2,3,-2,4]

Expected Output: 6

Constraints: $1 \leq n \leq 2 \times 10^4$

Q12. Subarray Sum Equals K

Count contiguous subarrays whose sum equals k.

Example: nums=[1,1,1], k=2

Expected Output: 2

Constraints: $1 \leq n \leq 2 \times 10^4$

Pattern/Topic: Prefix Sum + HashMap

Strings

Q13. Reverse a String

Reverse a string without using a built-in reverse method.

Example: "hello"

Expected Output: "olleh"

Constraints: $1 \leq \text{length} \leq 10^5$

Pattern/Topic: Character Array

Q14. Palindrome String

Check whether a string is a palindrome, ignoring letter case.

Example: "Madam"

Expected Output: true

Constraints: $1 \leq \text{length} \leq 10^5$

Pattern/Topic: Two Pointer

Q15. Character Frequency

Count the frequency of every character in a string.

Example: "banana"

Expected Output: b=1, a=3, n=2

Constraints: $1 \leq \text{length} \leq 10^5$

Pattern/Topic: Frequency Array

Q16. First Non-Repeating Character

Return the index of the first character that occurs once, or -1 if none exists.

Example: "leetcode"

Expected Output: 0

Constraints: $1 \leq \text{length} \leq 10^5$

Pattern/Topic: Frequency

Q17. Valid Anagram

Determine whether two strings contain exactly the same character frequencies.

Example: "listen", "silent"

Expected Output: true

Constraints: $1 \leq \text{lengths} \leq 10^5$

Pattern/Topic: Frequency

Q18. Frequency Sort

Sort characters in decreasing order of their frequency. Equal-frequency characters may be in any order.

Example: "tree"

Expected Output: "eert" or "eetr"

Constraints: $1 \leq \text{length} \leq 10^5$

Pattern/Topic: HashMap + Sorting

Q19. Longest Substring Without Repeating Characters

Return the length of the longest substring containing no repeated characters.

Example: "abcabcbb"

Expected Output: 3

Constraints: $1 \leq \text{length} \leq 10^5$

Pattern/Topic: Sliding Window

Q20. Longest Substring with At Most K Distinct Characters

Return the length of the longest substring containing at most k distinct characters.

Example: "eceba", k=2

Expected Output: 3

Constraints: $1 \leq \text{length} \leq 10^5$

Pattern/Topic: Sliding Window

Q21. Longest Substring with Exactly K Unique Characters

Return the length of the longest substring containing exactly k distinct characters, or -1 if none exists.

Example: "aabacbebebe", k=3

Expected Output: 7

Constraints: $1 \leq \text{length} \leq 10^5$

Pattern/Topic: Sliding Window

Q22. Find All Anagrams

Return all starting indices in s where an anagram of p occurs.

Example: s="cbaebabacd", p="abc"

Expected Output: [0,6]

Constraints: $1 \leq |p| \leq |s| \leq 10^5$

Pattern/Topic: Fixed Sliding Window

Q23. Maximum Sum of Subarray of Size K

Find the maximum sum among all contiguous subarrays of size k.

Example: [2,1,5,1,3,2], k=3

Expected Output: 9

Constraints: $1 \leq k \leq n \leq 10^5$

Pattern/Topic: Fixed Window

Q24. Longest Subarray with At Most K Zeroes [Medium]

Given a binary array, find the longest subarray containing at most k zeroes.

Example: [1,1,1,0,0,0,1,1,1,0], k=2

Expected Output: 6

Constraints: $1 \leq n \leq 10^5$

Pattern/Topic: Variable Window

Q25. Longest Substring with At Most K Distinct

Return the length of the longest substring containing at most k distinct characters.

Example: "aaabbcc", k=2

Expected Output: 5

Constraints: $1 \leq \text{length} \leq 10^5$

Pattern/Topic: Variable Window

Q26. Subarrays with At Most K Distinct

Count contiguous subarrays containing at most k distinct integers.

Example: [1,2,1,2,3], k=2

Expected Output: 12

Constraints: $1 \leq n \leq 10^5$

Pattern/Topic: Sliding Window

Q27. Longest Repeating Character Replacement

Find the longest substring that can be changed into one repeated uppercase character using at most k replacements.

Example: "AABABBA", $k=1$

Expected Output: 4

Constraints: $1 \leq \text{length} \leq 10^5$

Pattern/Topic: Sliding Window

Q28. Minimum Window Substring

Find the smallest substring of s containing all characters of t with required frequencies. Return "" if impossible.

Example: $s="ADOBECODEBANC"$, $t="ABC"$

Expected Output: "BANC"

Constraints: $1 \leq |t| \leq |s| \leq 10^5$

Pattern/Topic: Sliding Window

Q29. Minimum Size Subarray Sum

Given positive integers, return the minimum length of a subarray whose sum is at least target; return 0 if none exists.

Example: $\text{target}=7$, $\text{nums}=[2,3,1,2,4,3]$

Expected Output: 2

Constraints: $1 \leq n \leq 10^5$

Pattern/Topic: Variable Window

Q30. Intersection of Two Arrays

Return the distinct values present in both arrays.

Example: $[1,2,2,1]$, $[2,2]$

Expected Output: $[2]$

Constraints: $1 \leq \text{lengths} \leq 10^5$

Pattern/Topic: HashSet

Q31. Union of Two Arrays [Easy]

Return the distinct values present in either array.

Example: $[1,2,3]$, $[2,3,4]$

Expected Output: $[1,2,3,4]$

Constraints: $1 \leq \text{lengths} \leq 10^5$

Pattern/Topic: HashSet

Q32. Equal 0s and 1s

Find the length of the longest contiguous subarray containing equal numbers of 0 and 1.

Example: [0,1,0,1,1,1,0]

Expected Output: 4

Constraints: $1 \leq n \leq 10^5$

Q33. Pairs with Given Difference

Count distinct value-pairs whose absolute difference is k.

Example: [1,5,3,4,2], k=2

Expected Output: 3

Constraints: $1 \leq n \leq 10^5$

Pattern/Topic: HashSet

Q34. 3Sum

Find all unique triplets whose sum is zero.

Example: [-1,0,1,2,-1,-4]

Expected Output: [[-1,-1,2],[-1,0,1]]

Constraints: $3 \leq n \leq 3000$

Pattern/Topic: Sorting + Two Pointer

Recursion

Q35. Factorial

Calculate n! using recursion.

Example: n=5

Expected Output: 120

Constraints: $0 \leq n \leq 12$ for int result

Pattern/Topic: Recursion

Q36. Recursive Array Sum [Easy]

Find the sum of all array elements using recursion.

Example: [1,2,3,4]

Expected Output: 10

Constraints: $1 \leq n \leq 10^4$

Pattern/Topic: Recursion

Q37. Recursive Maximum

Find the maximum array element using recursion.

Example: [3,7,2,9,5]

Expected Output: 9

Constraints: $1 \leq n \leq 10^4$

Pattern/Topic: Recursion

Q38. Reverse String Recursively [Easy]

Reverse a string using recursion.

Example: "abcd"

Expected Output: "dcba"

Constraints: $1 \leq \text{length} \leq 10^4$

Pattern/Topic: Recursion

Q39. Recursive Palindrome [Medium]

Check whether a string is a palindrome using recursion.

Example: "racecar"

Expected Output: true

Constraints: $1 \leq \text{length} \leq 10^4$

Pattern/Topic: Recursion

Q40. Power Using Recursion [Medium]

Calculate x^n using an efficient recursive approach.

Example: $x=2, n=10$

Expected Output: 1024

Constraints: $0 \leq n \leq 30$

Pattern/Topic: Divide and Conquer

Q41. Generate Subsequences

Generate all subsequences of a string while preserving character order.

Example: "abc"

Expected Output: ["", "a", "b", "ab", "c", "ac", "bc", "abc"]

Constraints: $1 \leq \text{length} \leq 20$

Pattern/Topic: Recursion / Include-Exclude

Q42. Generate All Subsets

Given distinct integers, generate every possible subset using recursion and backtracking.

Example: [1,2,3]

Expected Output: 8 subsets

Constraints: $1 \leq n \leq 15$

Pattern/Topic: Backtracking

Q43. Permutations of Array

Generate all permutations of an array of distinct integers.

Example: [1,2,3]

Expected Output: 6 permutations

Constraints: $1 \leq n \leq 9$

Pattern/Topic: Backtracking

Q44. Permutations of String

Generate all permutations of a string containing distinct characters.

Example: "ABC"

Expected Output: 6 permutations

Constraints: $1 \leq \text{length} \leq 9$

Pattern/Topic: Backtracking

Q45. Combinations of K Numbers [Medium]

Generate all combinations of k numbers chosen from 1 through n.

Example: n=4, k=2

Expected Output: [[1,2],[1,3],[1,4],[2,3],[2,4],[3,4]]

Constraints: $1 \leq k \leq n \leq 20$

Pattern/Topic: Backtracking

Q46. Target Sum Combinations

Given positive candidates that may be reused, find all combinations summing to target.

Example: [2,3,6,7], target=7

Expected Output: [[2,2,3],[7]]

Constraints: $1 \leq n \leq 30$; target ≤ 40

Pattern/Topic: Backtracking

Q47. Combination Sum [Medium]

Given distinct positive candidates and a target, return all unique combinations. Each candidate can be used unlimited times.

Example: [2,3,6,7], target=7

Expected Output: [[2,2,3],[7]]

Constraints: $1 \leq n \leq 30$; target ≤ 40

Pattern/Topic: Backtracking

Q48. Combination Sum II

Candidates may contain duplicates. Return unique combinations summing to target; each element can be used once.

Example: [10,1,2,7,6,1,5], target=8

Expected Output: [[1,1,6],[1,2,5],[1,7],[2,6]]

Constraints: $1 \leq n \leq 100$; target ≤ 30

Pattern/Topic: Backtracking + Duplicate Handling

Q49. Generate Parentheses

Generate all well-formed combinations of n pairs of parentheses.

Example: n=3

Expected Output: ["((()))","(()())","(())()","()(())","()()()"]

Constraints: $1 \leq n \leq 8$

Pattern/Topic: Backtracking

Q50. Rat in a Maze

Given an $n \times n$ binary maze, find all paths from top-left to bottom-right without revisiting cells.

Example: 4×4 binary maze

Expected Output: All valid paths

Constraints: $1 \leq n \leq 10$

Pattern/Topic: Backtracking

Q51. N-Queens

Place n queens so no two queens attack each other. Return all valid arrangements.

Example: $n=4$

Expected Output: 2 solutions

Constraints: $1 \leq n \leq 10$

Pattern/Topic: Backtracking

Q52. Sudoku Solver

Solve a partially filled 9×9 Sudoku board so every row, column, and 3×3 box contains digits 1-9 exactly once.

Example: Standard Sudoku board

Expected Output: Completed valid board

Constraints: 9×9 board

Pattern/Topic: Backtracking

Q53. Traverse a Linked List

Given the head of a singly linked list, print all elements in order from the first node to the last node.

Example: $1 \rightarrow 2 \rightarrow 3 \rightarrow 4 \rightarrow 5$

Expected Output: $1\ 2\ 3\ 4\ 5$

Constraints: $0 \leq n \leq 10^5$

Pattern/Topic: Linked List Traversal

Q54. Insert Node at Beginning

Given the head of a singly linked list and a value, insert a new node at the beginning and return the new head.

Example: $1 \rightarrow 2 \rightarrow 3$, value = 0

Expected Output: $0 \rightarrow 1 \rightarrow 2 \rightarrow 3$

Constraints: $0 \leq n \leq 10^5$

Pattern/Topic: Singly Linked List / Insertion

Q55. Insert Node at End

Given the head of a singly linked list and a value, insert a new node at the end of the list.

Example: $1 \rightarrow 2 \rightarrow 3$, value = 4

Expected Output: $1 \rightarrow 2 \rightarrow 3 \rightarrow 4$

Constraints: $0 \leq n \leq 10^5$

Pattern/Topic: Singly Linked List / Insertion

Q56. Delete Node by Value

Given the head of a singly linked list and a target value, delete the first node containing that value. If the value does not exist, leave the list unchanged.

Example: $1 \rightarrow 2 \rightarrow 3 \rightarrow 4$, target = 3

Expected Output: $1 \rightarrow 2 \rightarrow 4$

Constraints: $0 \leq n \leq 10^5$

Pattern/Topic: Singly Linked List / Deletion

Q57. Reverse a Linked List

Reverse a singly linked list and return the new head.

Example: $1 \rightarrow 2 \rightarrow 3 \rightarrow 4 \rightarrow 5$

Expected Output: $5 \rightarrow 4 \rightarrow 3 \rightarrow 2 \rightarrow 1$

Constraints: $0 \leq n \leq 10^5$

Pattern/Topic: Iterative Pointer Manipulation

Q58. Find Middle of Linked List

Given the head of a singly linked list, return the middle node. If the list has an even number of nodes, return the second middle node.

Example: $1 \rightarrow 2 \rightarrow 3 \rightarrow 4 \rightarrow 5$

Expected Output: 3

Constraints: $1 \leq n \leq 10^5$

Pattern/Topic: Fast and Slow Pointer

Q59. Detect Cycle in Linked List

Determine whether a singly linked list contains a cycle.

Example: $1 \rightarrow 2 \rightarrow 3 \rightarrow 4$, where $4.next \rightarrow 2$

Expected Output: true

Constraints: $0 \leq n \leq 10^5$

Pattern/Topic: Floyd's Cycle Detection

Q60. Find Starting Node of Cycle

Given a singly linked list that may contain a cycle, return the node where the cycle begins. Return null if no cycle exists.

Example: $1 \rightarrow 2 \rightarrow 3 \rightarrow 4 \rightarrow 5$, where $5.next \rightarrow 3$

Expected Output: Node containing 3

Constraints: $0 \leq n \leq 10^5$

Pattern/Topic: Floyd's Cycle Detection

Q61. Remove Nth Node from End

Remove the nth node from the end of a singly linked list and return the modified list.

Example: $1 \rightarrow 2 \rightarrow 3 \rightarrow 4 \rightarrow 5$, n = 2

Expected Output: $1 \rightarrow 2 \rightarrow 3 \rightarrow 5$

Constraints: $1 \leq n \leq \text{length} \leq 10^5$

Pattern/Topic: Two Pointer / Fast and Slow Pointer

Q62. Merge Two Sorted Linked Lists

Given two sorted singly linked lists, merge them into one sorted linked list.

Example: $1 \rightarrow 3 \rightarrow 5$ and $2 \rightarrow 4 \rightarrow 6$

Expected Output: $1 \rightarrow 2 \rightarrow 3 \rightarrow 4 \rightarrow 5 \rightarrow 6$

Constraints: $0 \leq n, m \leq 10^5$

Pattern/Topic: Linked List / Two Pointer

Q63. Insert Node at Beginning of Doubly Linked List

Given the head of a doubly linked list and a value, insert a new node at the beginning while maintaining both **next** and **prev** pointers correctly.

Example: $1 \rightleftharpoons 2 \rightleftharpoons 3$, value = 0

Expected Output: $0 \rightleftharpoons 1 \rightleftharpoons 2 \rightleftharpoons 3$

Constraints: $0 \leq n \leq 10^5$

Pattern/Topic: Doubly Linked List / Insertion

Q64. Insert Node at End of Doubly Linked List

Insert a new node at the end of a doubly linked list while correctly maintaining the **next** and **prev** pointers.

Example: $1 \rightleftharpoons 2 \rightleftharpoons 3$, value = 4

Expected Output: $1 \rightleftharpoons 2 \rightleftharpoons 3 \rightleftharpoons 4$

Constraints: $0 \leq n \leq 10^5$

Pattern/Topic: Doubly Linked List / Insertion

Q65. Delete a Given Node from Doubly Linked List

Given a reference to a node in a doubly linked list, delete that node without traversing the list from the head.

Example: $1 \rightleftharpoons 2 \rightleftharpoons 3 \rightleftharpoons 4$, delete node 3

Expected Output: $1 \rightleftharpoons 2 \rightleftharpoons 4$

Constraints: $1 \leq n \leq 10^5$

Pattern/Topic: Doubly Linked List / Pointer Manipulation

Q66. Reverse a Doubly Linked List

Reverse a doubly linked list by correctly swapping the **next** and **prev** relationships of every node.

Example: $1 \rightleftharpoons 2 \rightleftharpoons 3 \rightleftharpoons 4$

Expected Output: $4 \rightleftharpoons 3 \rightleftharpoons 2 \rightleftharpoons 1$

Constraints: $0 \leq n \leq 10^5$

Pattern/Topic: Doubly Linked List / Reversal

Q67. Find a Pair with Given Sum

Given a sorted doubly linked list and a target value, determine whether there exists a pair of nodes whose values add up to the target.

Example: $1 \rightleftharpoons 2 \rightleftharpoons 4 \rightleftharpoons 5 \rightleftharpoons 6$, target = 7

Expected Output: $[1, 6]$

Constraints: $1 \leq n \leq 10^5$

Pattern/Topic: Two Pointer / Doubly Linked List

Q68. Remove All Occurrences of a Value

Given a doubly linked list and a target value, remove every node whose value equals the target.

Example: $1 \rightleftharpoons 2 \rightleftharpoons 2 \rightleftharpoons 3 \rightleftharpoons 2$, target = 2

Expected Output: $1 \rightleftharpoons 3$

Constraints: $0 \leq n \leq 10^5$

Pattern/Topic: Doubly Linked List / Deletion

Q69. Check Whether a Doubly Linked List is a Palindrome

Determine whether the sequence of values in a doubly linked list reads the same from both directions.

Example: $1 \rightleftharpoons 2 \rightleftharpoons 3 \rightleftharpoons 2 \rightleftharpoons 1$

Expected Output: `true`

Constraints: $0 \leq n \leq 10^5$

Pattern/Topic: Two Pointer / Doubly Linked List

Q70. Traverse a Circular Linked List

Given the head of a circular singly linked list, print every node exactly once without entering an infinite loop.

Example: $1 \rightarrow 2 \rightarrow 3 \rightarrow 4 \rightarrow \text{back to } 1$

Expected Output: `1 2 3 4`

Constraints: $0 \leq n \leq 10^5$

Pattern/Topic: Circular Linked List / Traversal

Q71. Insert Node at Beginning of Circular Linked List

Insert a new node at the beginning of a circular singly linked list while maintaining the circular connection.

Example: $1 \rightarrow 2 \rightarrow 3 \rightarrow \text{back to } 1$, value = 0

Expected Output: $0 \rightarrow 1 \rightarrow 2 \rightarrow 3 \rightarrow \text{back to } 0$

Constraints: $0 \leq n \leq 10^5$

Pattern/Topic: Circular Linked List / Insertion

Q72. Insert Node at End of Circular Linked List

Insert a new node at the end of a circular singly linked list and maintain the circular connection.

Example: $1 \rightarrow 2 \rightarrow 3 \rightarrow \text{back to } 1$, value = 4

Expected Output: $1 \rightarrow 2 \rightarrow 3 \rightarrow 4 \rightarrow \text{back to } 1$

Constraints: $0 \leq n \leq 10^5$

Pattern/Topic: Circular Linked List / Insertion

Q73. Delete a Node from Circular Linked List

Given the head of a circular linked list and a target value, delete the first node containing the target. Handle deletion of the head and the only node correctly.

Example: $1 \rightarrow 2 \rightarrow 3 \rightarrow 4 \rightarrow \text{back to } 1$, target = 1

Expected Output: $2 \rightarrow 3 \rightarrow 4 \rightarrow \text{back to } 2$

Constraints: $0 \leq n \leq 10^5$

Pattern/Topic: Circular Linked List / Deletion

Q74. Detect Whether a Linked List is Circular

Given a linked list, determine whether following **next** pointers eventually returns to the head node.

Example: **1 → 2 → 3 → 4 → back to 1**

Expected Output: **true**

Constraints: $0 \leq n \leq 10^5$

Pattern/Topic: Circular Linked List / Cycle Detection

Q75. Split Circular Linked List into Two Halves

Given a circular linked list, split it into two circular linked lists of approximately equal size. If the number of nodes is odd, the first list should contain one extra node.

Example: **1 → 2 → 3 → 4 → 5 → back to 1**

Expected Output: **1 → 2 → 3 → back to 1** and **4 → 5 → back to 4**

Constraints: $1 \leq n \leq 10^5$

Pattern/Topic: Fast and Slow Pointer / Circular Linked List